Frozen Foundation-Model Features for ILC vs IDC Classification on TCGA-BRCA

Carmel Prosper Sagbo
Carnegie Mellon University Africa
Upanzi – Digital Public Infrastructure Network
Kigali, Rwanda
csagbo@andrew.cmu.edu

Abstract

Invasive lobular carcinoma (ILC) and invasive ductal carcinoma (IDC) require different clinical management, but ILC's diffuse, single-cell infiltrative growth pattern makes it harder to classify from histology than IDC's cohesive nests. We evaluate whether *frozen* pathology foundation-model features can match the accuracy of fully fine-tuned, task-specific CNNs on this classification task, using TCGA-BRCA whole-slide images ($n = 109$ patients, 55 ILC / 54 IDC, site-grouped 5-fold cross-validation to prevent scanner/batch leakage). We extract frozen tile features from UNI2-h (1536-dim) and a baseline ImageNet-pretrained ResNet-18 (512-dim), and train 10 classifier-head families on top (logistic regression, SVM, MLP, gated-attention MIL, plain attention, graph attention, and four GAT composites), swept across 172 total configurations including a 152-point hyperparameter ablation (dropout, optimizer, scheduler, epochs). Our best configuration (UNI2-h + gated-attention MIL) reaches patient-level AUROC $= 0.920$, matching the external benchmark of Ektefaie et al. [2021] – a fully fine-tuned Inception-v3 CNN achieving AUROC $= 0.920$ on the same dataset and task – despite using zero gradient updates to the backbone. UNI2-h features outperform the ResNet-18 baseline on every one of 86 matched configurations (median AUROC 0.894 vs. 0.701). We also report two negative results: a naive mean-ensemble of correlated classifier heads underperforms its best single member, and a zero-shot LLM baseline underperforms a trivial sign-vote control on the same inputs.

1 Introduction

Distinguishing invasive lobular carcinoma (ILC, $\sim$10–15% of invasive breast cancers) from invasive ductal carcinoma (IDC, $\sim$70–80%) from H&E histology is clinically important: the two subtypes differ in growth pattern, metastatic routes, and response to neoadjuvant therapy, but ILC's single-cell "Indian-file" infiltration pattern lacks the cohesive architecture that makes IDC comparatively easy to recognize, and is a frequent source of diagnostic disagreement among pathologists.

Deep learning on whole-slide images (WSIs) has reached strong performance on this task, but historically required full end-to-end fine-tuning of a CNN or transformer backbone on thousands of tiles per slide – expensive in compute, and in curated labeled data. Pathology foundation models (UNI, UNI2-h, CONCH, Virchow, Phikon) pretrained via self-supervision on large, diverse slide corpora offer a cheaper alternative: *frozen* feature extraction followed by training only a lightweight classifier head. Whether this frozen-feature approach can *match* task-specific fine-tuning, rather than only approach it, is the question this work addresses directly, with a systematic sweep across classifier architectures and hyperparameters rather than a single reported configuration.

Preprint. Under review.

This evaluation is also motivated by a concrete, practical goal: building reproducible, well-documented evidence of foundation-model experience on a real pathology classification task, as preparation for a liver pathology fellowship application that was previously declined for lacking exactly this kind of hands-on evidence.

2 Related Work

Ektefaie et al. [2021], *npj Breast Cancer*, is the directly comparable external benchmark. They fully fine-tuned an ImageNet-pretrained Inception-v3 CNN on the top 200 tiles/slide (ranked by tissue density) from TCGA-BRCA, with patient-level prediction via averaging tile predictions, and splits grouped by tissue source site – the same site-leakage safeguard used in this work. They report patient-level AUC $= 0.920$ (95% CI 0.845–0.993; tile-level AUC $= 0.800$) for lobular-vs-ductal classification, on a cohort of $\sim$1,099 TCGA-BRCA patients ($\sim$22% lobular; held-out test split size for this specific task not stated).

Zhang [2025], arXiv:2508.14779 (ISBI 2026 submission), studies hospital-source bias in frozen UNI/CONCH features and reports disease AUC $= 0.98 \pm 0.01$ for a UNI + MLP baseline on IDC vs ILC – but this number is not a fair comparison target: it is *patch-level* rather than patient-level, computed on only 4,029 patches pre-filtered by CONCH zero-shot to keep high-confidence (easy) patches, from a small cohort (80 WSIs, 4 hospitals) restricted to one demographic group (White female patients aged 60–79). We report it here only for completeness, and separately because the paper's own finding – hospital/site is strongly predictable from frozen features (AUC $\sim$0.95–0.96) – reinforces the importance of the site-grouped cross-validation used throughout this work.

Foundation models. UNI2-h (Mahmood Lab) is a ViT-H/14 vision transformer pretrained via self-supervision on a large multi-site histopathology corpus; we use it here strictly as a frozen feature extractor, with no fine-tuning of backbone weights.

3 Methods

Cohort. 110 TCGA-BRCA patients (55 ILC, 55 IDC) with a usable diagnostic (Primary Tumor, not Metastatic) H&E slide, balanced per tissue-source site; one slide lacked scanner MPP metadata and was excluded (not imputed), leaving $n = 109$ (55 ILC / 54 IDC) across 16 both-class sites.

Tiling and feature extraction. Slides were resampled to a common 0.5 µm/pixel (the smallest target requiring only downsampling across the full cohort), tiled at 224×224px (up to 1,000 tissue tiles/slide by Otsu-threshold tissue fraction), and passed frozen through two backbones: **UNI2-h** (ViT-H/14, 1536-dim) and a **baseline** ImageNet-pretrained ResNet-18 (512-dim, penultimate layer). Both backbones see identical tile coordinates per slide.

Cross-validation. Site-grouped 5-fold CV (no tissue-source site split across folds, both classes present in every fold, verified by assertion) repeated across 5 seeds; each patient's final prediction averages its out-of-fold probability across seeds. 95% CIs via patient-level bootstrap (2,000 resamples).

Classifier heads. (1) Logistic regression and (2) SVM (4 kernels) on mean-pooled tile features; (3) a small MLP; (4) gated-attention MIL [Ilse et al., 2018]; (5) plain (ungated) attention pooling; (6) graph attention (GAT, Veličković et al. 2018) over a spatial k-NN tile graph; (7–10) four GAT composites (GAT backbone $\rightarrow$ LR / SVM / MLP / MIL as the final classifier, the LR/SVM variants trained via a differentiable proxy head since those classifiers are not themselves differentiable). (11) An exploratory zero-shot LLM baseline: each held-out patient's embedding is reduced to 8 class-oriented signed z-scores on the most train-discriminative feature dimensions, and a small local instruction-tuned LLM (Qwen2.5-0.5B-Instruct, no external API) is asked to classify from those signals, compared against a trivial no-LLM sign-majority-vote control.

Hyperparameter ablation. For MLP and MIL heads (both feature sets): dropout $\times\{0.1, 0.2, 0.3\}$ $\times$ epochs $\{50, 100, 150\}$ $\times$ optimizer-profile {Adam@3e-4, SGD@0.01} $\times$ scheduler {cosine, step} $= 36$ combinations $\times\, 4$ base configs $= 144$ runs. SVM kernel ablation: {linear, rbf, poly, sigmoid} $\times\, 2$ feature sets $= 8$ runs. Combined with 20 base architecture configs and 2 LLM configs: **172 total completed configurations**.

4 Results

The best configuration, UNI2-h + gated-attention MIL, reaches patient-level AUROC $= 0.920$ – matching Ektefaie et al. [2021]'s fully fine-tuned Inception-v3 benchmark exactly, with zero backbone gradient updates.

Table 1: Top 10 configurations by patient-level AUROC.

Feature	Head	AUROC	Balanced Acc.	Brier
UNI2-h	MIL (dropout 0.2, Adam, step, 50ep)	0.920	0.817	0.127
UNI2-h	MIL (dropout 0.3, Adam, cosine, 50ep)	0.920	0.826	0.124
UNI2-h	MIL (dropout 0.1, SGD, step, 50ep)	0.919	0.854	0.114
UNI2-h	MIL (dropout 0.1, Adam, cosine, 50ep)	0.919	0.835	0.126
UNI2-h	Attention (plain, base config)	0.919	0.844	0.123
UNI2-h	MIL (dropout 0.3, Adam, step, 50ep)	0.919	0.826	0.126
UNI2-h	MIL (dropout 0.1, Adam, cosine, 150ep)	0.918	0.826	0.132
UNI2-h	MIL (dropout 0.1, Adam, step, 50ep)	0.918	0.826	0.127
UNI2-h	MIL (base config)	0.917	0.835	0.125
UNI2-h	MIL (dropout 0.3, Adam, step, 100ep)	0.917	0.826	0.123

Table 2: UNI2-h vs. baseline CNN, across 86 matched configurations per feature set.

	UNI2-h	Baseline ResNet-18
Median AUROC	0.894	0.701
Max AUROC	0.920	0.762

UNI2-h's median performance (0.894) exceeds the baseline's maximum (0.762) – a complete separation with no overlap across 172 configurations.

5 Discussion

MIL and plain attention converge to the same ceiling. Across all dropout/optimizer/scheduler combinations, both gated-attention MIL and plain (ungated) attention independently plateau at 0.917–0.920 on UNI2-h. This suggests the aggregation mechanism's gating component is not the binding constraint on this task – a more sophisticated pooling mechanism is unlikely to help without more training data (Section 6).

Graph attention underperforms plain attention. GAT and its composites reach only 0.88–0.89, below plain attention's 0.919. A naive spatial k-NN graph over tile coordinates does not capture useful structure beyond what unordered attention already extracts – the explicit graph inductive bias is not earning its complexity here.

SGD fails specifically on weaker features. `baseline_cnn + MLP + SGD` produced AUROC as low as 0.44 (below chance) in four separate hyperparameter combinations, while the identical architecture with Adam reaches 0.75–0.76, and `uni2h + MLP + SGD` trains normally (0.86–0.87). This is a repeatable optimizer$\times$feature-quality interaction, not noise: SGD's lack of per-parameter adaptive scaling appears to need the richer UNI2-h feature space to find a good basin reliably.

A naive ensemble did not help. Averaging the MIL model's predictions with three MLP variants (individually 0.87–0.90 AUROC) produced an ensemble AUROC of 0.900 – *below* the best single member (0.911 in that comparison). Because all members shared the same UNI2-h feature space, their errors were likely correlated, and mean-averaging a strong model with weaker ones pulled the average down rather than reducing variance. This motivates cross-foundation-model ensembling (genuinely diverse feature spaces) over same-feature cross-head ensembling as the more promising direction.

The LLM baseline underperforms its own trivial control. The zero-shot LLM classifier reached AUROC $= 0.568$, below the no-LLM sign-vote control (0.670) on the identical input signals. A small instruction-tuned LLM does not reliably aggregate even a handful of pre-oriented numeric signals

better than a trivial majority vote – a clear negative result for this style of LLM-on-tabular-features approach at this scale.

6 Limitations and Future Work

Cohort size. $n = 109$ is small for deep learning even with frozen features, and is a balanced subsample of a larger eligible pool (757 IDC + 191 ILC patients with a usable diagnostic slide). That three architecturally different aggregation mechanisms (gated MIL, plain attention, GAT) all independently converge to the same $\sim$0.92 ceiling is consistent with sample size, not architecture, being the binding constraint. A cohort expansion to **348 patients** (174 ILC + 174 IDC, all available ILC matched per-site with IDC) is in progress at the time of writing, with site-grouped 5-fold feasibility already confirmed.

Single foundation model. Results reported here use only UNI2-h. Phikon-v2 and Virchow2 feature extraction is underway; cross-foundation-model ensembling (motivated by the within-feature-space ensemble's negative result in Section 5) is planned once available.

Frozen features only. We did not evaluate partial or LoRA fine-tuning of the UNI2-h backbone, which carries real overfitting risk at this sample size but could recover additional task-specific signal.

No stain normalization. Cross-site H&E stain and scanner variation was not explicitly corrected for beyond the site-grouped CV safeguard; this is a known confound in multi-institutional cohorts like TCGA.

7 Conclusion

A frozen UNI2-h foundation model with a lightweight gated-attention MIL head matches a fully fine-tuned, task-specific Inception-v3 CNN [Ektefaie et al., 2021] on ILC vs IDC classification from TCGA-BRCA whole-slide images – patient-level AUROC 0.920 on both – at a fraction of the training cost, across a systematic sweep of 172 configurations rather than a single favorable run. The result is robust to classifier-head choice (MIL and plain attention converge identically) and to hyperparameter choice (stable across dropout, optimizer, and schedule), while graph-structured attention, naive same-feature ensembling, and a zero-shot LLM baseline each underperformed simpler alternatives – negative results we report alongside the positive one. Cohort expansion to 348 patients and cross-foundation-model ensembling are the next steps toward improving on this ceiling.

References

Yasha Ektefaie, Wei Yuan, Deborah A. Dillon, Nancy U. Lin, Jeffrey A. Golden, Isaac S. Kohane, and Kun-Hsing Yu. Integrative multiomics-histopathology analysis for breast cancer classification. *npj Breast Cancer*, 7:147, 2021. doi: 10.1038/s41523-021-00357-y.

Maximilian Ilse, Jakub M. Tomczak, and Max Welling. Attention-based deep multiple instance learning. In *Proceedings of the 35th International Conference on Machine Learning*, 2018.

Petar Veličković, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. Graph attention networks. In *International Conference on Learning Representations*, 2018.

Mengliang Zhang. Hospital-specific bias in patch-based pathology models. *arXiv preprint arXiv:2508.14779*, 2025.